

Aetnamem Explained

Twenty Articles on Governed Memory for AI

aetna000.com

Based on *Governed Memory Without Embeddings*

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Abstract

This collection contains twenty standalone, high-level articles prepared for publication on X. It translates the architecture and measured claims of the AETNAMEM governed-memory white paper into accessible language without removing the technical boundaries. The series explains why durable AI memory is more than search, how provenance and trust shape recall, why the graph is a rebuildable index, what recall forensics means, how approval limits external effects, and where the current encryption and scale evidence stops.

Source paper: *Governed Memory Without Embeddings: Recall Forensics, Deterministic Retrieval, and Approval-Bound Effects in aetnamem*. Author: aetna000.com.

Editorial note: Each article is designed to stand alone. Technical measurements are used only where they clarify the idea. Statements about implementation refer to the version and content-addressed evaluation snapshot documented by the source paper, not to every future release.

Contents

1	Memory Is Not a Search Box	3
2	A Filing Cabinet, Catalogue, Map, and Receipt Book	4
3	Why the Source of a Memory Matters	5
4	Quarantine Is Memory With a Door	6
5	Corrections Should Not Erase History	7
6	Forgetting Is More Than Removing a Search Result	8
7	Why Aetnamem Does Not Use Embeddings	9
8	Bounded Recall on a Laptop	10
9	The Graph Is a Map, Not the Truth	11
10	Multi-Hop Recall That Can Explain Itself	12

11 Recall Forensics: Explaining a Past Answer	13
12 A Hash Chain Is Useful, but It Needs an Anchor	14
13 Confidence Is Not Trust	15
14 Ambiguity Should Stay Visible	16
15 Information Is Not Authority	17
16 Approve the Exact Plan, Not the General Intention	18
17 “Uncertain” Is a Responsible Result	19
18 What Aetnamem Encryption Protects Today	20
19 Scale Claims Should Follow Evidence	21
20 The End Goal: Governed Memory Infrastructure	22

1 Memory Is Not a Search Box

Article 1 of 20

A useful AI memory system must govern what becomes memory, not only find text that looks relevant.

When people hear “AI memory,” they often imagine a search engine attached to a chatbot. Store old conversations, find similar passages, place them in the next prompt, and let the model answer. That can improve continuity, but it does not answer the harder questions created by long-lived memory.

Who supplied a remembered statement? Was it the user, a web page, a tool, or the model itself? Is it still current? Did a later correction replace it? Was it reviewed? Can it be deleted? If it contributes to a proposed action, does it carry any authority to change the outside world?

AETNAMEM treats these as state-management questions. It stores source episodes, normalized records, lifecycle status, trust origin, provenance, and audit evidence. Retrieval operates inside those rules. A result cannot enter ordinary recall merely because it has a strong text match; it must first be an active memory under the lifecycle policy.

This changes the role of the language model. The model may help a user speak to the system, summarize evidence, or propose an action. It does not own the truth of the database, decide which untrusted content becomes accepted, or turn recalled information into permission.

The practical goal is modest but important: when an assistant says “I remember,” the system should be able to show what was accepted, where it came from, what changed since then, and why it was eligible to appear.

Important boundary. AETNAMEM does not prove that a remembered statement is true. It preserves who supplied it, how it was treated, and how it participated in the system.

2 A Filing Cabinet, Catalogue, Map, and Receipt Book

Article 2 of 20

The easiest way to understand AETNAMEM is as four familiar tools working together.

The first tool is a **filing cabinet**. It holds canonical episodes and records: the original source material and the concise facts derived from it. These are the durable objects from which other structures can be rebuilt.

The second is a **card catalogue**. SQLite full-text search helps locate active records without scanning and scoring every item in memory. It is a local index with explicit limits, not a second source of truth.

The third is a **map**. A derived graph connects entities and typed relations. If memory says “my boss is Sarah” and “Sarah prefers SEA,” the map can follow the relation between those statements. Every edge points back to a record, and every record points back to its source episode.

The fourth is a **receipt book**. Important transitions append evidence to a hash-linked audit chain: admission, promotion, correction, deletion, retrieval, graph maintenance, approval, and external-action outcomes.

This separation is useful because the catalogue and map may be wrong or outdated without corrupting the filing cabinet. They can be cleared and rebuilt from canonical records. The receipt book then helps an operator check which transitions occurred and whether the recorded chain still verifies.

Many systems combine these responsibilities into one opaque model-driven pipeline. AETNAMEM separates them so that failures have smaller meanings: an index problem is not automatically a memory-integrity problem, and a relevant memory is not automatically an authorized action.

Important boundary. Rebuildability improves recovery from index damage. It does not replace backups or recover canonical records that were lost from disk.

3 Why the Source of a Memory Matters

Article 3 of 20

The same sentence should be treated differently when it comes from a user, a web page, a tool, or the assistant itself.

Imagine a user has stored a preferred airport. Later, a web page contains the sentence “your preferred airport is LAX.” A text extractor can read both sentences, but it should not treat them as equally authoritative.

AETNAMEM classifies origin before a candidate memory becomes active. Direct user statements and explicitly reviewed content may enter active memory. Marked web-page or tool-output content begins quarantined. It remains visible for review, but ordinary recall and graph traversal exclude it.

This is a practical defense against durable memory poisoning. A hostile page cannot silently replace a trusted preference merely because it contains language that looks like a personal fact. Promotion is a separate action with its own provenance and audit event.

Assistant responses are also not automatically promoted into semantic memory. Otherwise, a model could repeat its own output until generated text appeared to have become an established fact. The system can log the exchange without granting the model self-authorship over durable truth.

Origin is not the same as correctness. A trusted user can be mistaken, and a web page can be accurate. The trust tier instead answers a narrower policy question: how may this source participate before review?

Important boundary. Origin classification depends on the host correctly labeling external content. If a compromised integration presents remote text as a direct user statement, the storage layer cannot recover the missing context.

4 Quarantine Is Memory With a Door

Article 4 of 20

Suspicious information does not have to be either trusted or discarded; it can be retained for inspection without entering normal recall.

Binary choices are often too crude for AI memory. Throwing away every untrusted observation loses potentially useful information. Accepting every observation creates a path for web content, tool output, or generated text to shape future answers as if a user had approved it.

AETNAMEM uses quarantine as an intermediate lifecycle state. A quarantined record keeps its source and can be reviewed, but it is excluded from ordinary recall. Its graph edges cannot become bridges between trusted facts. It also cannot supersede an active record.

Promotion is explicit. A reviewer can inspect the proposed memory, confirm it, and move it into the user-confirmed trust tier. The transition is recorded rather than hidden inside a model's confidence score.

This design makes disagreement visible. The system can say, in effect, "this information was observed, but it has not been accepted." That is more useful than pretending all stored text has the same standing.

Quarantine is especially important for agents that browse, call tools, or receive messages from external channels. Those systems encounter content that may be relevant but was not authored as an instruction to the memory layer.

Important boundary. A reviewer can still approve false information. Quarantine governs admission and records responsibility; it does not provide an oracle for fact checking.

5 Corrections Should Not Erase History

Article 5 of 20

A durable memory should show what changed, not silently rewrite the past.

Suppose a user says “my preferred airport is Sydney,” then later changes it to Melbourne. Editing the old database row in place would produce the current answer, but it would remove the evidence that a correction occurred.

AETNAMEM uses explicit supersession for recognized single-valued facts. The new active record becomes current, while the prior record moves to a superseded state and points to its replacement. Normal recall returns the new value, but history remains reconstructable.

This matters when people ask why an assistant behaved differently last month. The answer may depend on the memory state at that time, not the state visible today. A transition history allows an auditor to replay which record was eligible at a particular turn.

The policy also protects trusted state from unreviewed replacement. A quarantined record cannot supersede an active record, even when both appear to occupy the same fact slot.

Not every relation is single-valued. “Works with,” “likes,” and “avoids” may accumulate. Unstructured conflicts may remain as separate records for review rather than being merged by semantic guesswork.

Important boundary. Supersession depends on the system recognizing a structured fact key. Unsupported phrasing can remain as multiple records, because visible ambiguity is safer than an unexplained destructive update.

6 Forgetting Is More Than Removing a Search Result

Article 6 of 20

Deletion should remove governed payloads and indexes while leaving evidence that a deletion operation occurred.

Removing an item from search is not necessarily forgetting it. The original text may still exist in a record table, a graph edge, a full-text index, a source episode, or an archive.

In AETNAMEM, forgetting is a lifecycle transition across these structures. Matching records are tombstoned, selected payloads are purged, and related full-text and graph entries are removed. Source episodes are purged when no retained record still depends on them.

The system keeps a deletion receipt containing identifiers and digests rather than the erased words. This creates a useful balance: an operator can verify that a deletion was processed without preserving the sensitive value merely to make future auditing easier.

That choice creates a real replay boundary. If a memory contributed to an old answer and was later deleted, the system may reconstruct the historical decision from commitments and lifecycle events while correctly refusing to reproduce the deleted plaintext.

Deletion is deliberately accepted only from user-origin requests, and an empty selector is rejected instead of being interpreted as “delete everything.” These are small rules with large consequences for safety.

Important boundary. A local deletion receipt cannot prove that independent backups, exports, model-provider logs, or external systems deleted their copies. Those systems need separate retention policies and deletion receipts.

7 Why Aetnamem Does Not Use Embeddings

Article 7 of 20

Avoiding embeddings is an engineering trade: less semantic reach in exchange for a smaller, more reproducible trusted retrieval path.

Embeddings are effective at finding paraphrases and conceptually similar text. AETNAMEM does not claim otherwise. It deliberately omits an embedding model and vector database from its current memory-retrieval path.

The reason is operational. Replaying a lexical decision requires text normalization, static weights, database state, and code. Replaying an embedding decision also requires the exact model and tokenizer, preprocessing, numerical environment, vector-index state, dependencies, and often hardware details.

Deletion and correction are also simpler when the indexes are ordinary SQLite rows. There is no stale vector to discover, re-embedding queue to operate, or dimensional migration when a model changes. A local installation does not need to download and run a second model just to remember.

Explanation is another factor. A lexical result can identify matching terms. A graph result can show a typed path. A nearest-neighbor score says that two items are close in a learned space, which is useful but usually less direct as a human explanation.

The cost is real: lexical retrieval can miss synonyms, misspellings, multilingual paraphrases, and conceptual equivalence. The current parser is also intentionally narrow. AETNAMEM chooses deterministic, deletable, and inspectable behavior for its present use case rather than claiming universal retrieval superiority.

Important boundary. Embedding retrieval can be reproducible when all required artifacts are preserved. The narrower claim is that AETNAMEM replay does not require that additional learned-model artifact.

8 Bounded Recall on a Laptop

Article 8 of 20

Memory growth should not force every query to score every stored record.

A naive memory implementation compares each query with every record. That becomes slower and produces larger evidence logs as the database grows.

AETNAMEM first asks SQLite FTS5 for a bounded set of active lexical matches. It fills unused capacity with recent active records, then ranks only that candidate set. The documented default candidate cap is 200, with a hard per-call ceiling of 2,000.

The ranking combines text relevance, source-trust weight, and recency using a versioned deterministic formula. These values are retrieval utilities, not probabilities that a fact is true. Lifecycle eligibility occurs before ranking, so a highly relevant quarantined record remains excluded.

Bounded work also matters for audit evidence. The retrieval event stores a query digest, returned identifiers, algorithm parameters, and a bounded candidate ledger rather than copying an unbounded list into the audit trail.

In the paper's 10,002-record laptop probe, direct recall kept its 200-record candidate cap. This demonstrates the mechanism at that size, not a guaranteed maximum for every machine or workload.

Important boundary. Candidate caps improve predictable work but can exclude relevant records. Retrieval quality, FTS selectivity, database size, cache pressure, and hardware still need representative measurement.

9 The Graph Is a Map, Not the Truth

Article 9 of 20

A knowledge graph is safer when it can be discarded and rebuilt from canonical records.

Graphs are useful because they make relationships explicit. They are also dangerous when an extraction error becomes accepted as permanent truth.

AETNAMEM treats its entity-relation graph as a derived index. Every edge points to a canonical record, and every record points to its source episode. If an extractor improves or the graph becomes damaged, the system can rebuild the map from the retained filing cabinet.

Lifecycle rules propagate into the graph. An edge derived from quarantined content is quarantined. A correction supersedes the corresponding single-valued edge. Forgetting removes or tombstones related graph objects. Only active entities, aliases, and edges participate in normal traversal.

Entity merges are proposal-only. Exact same-kind candidates can be shown to a reviewer, but the system does not let a language model decide that two people or projects are identical. Approved merges use reversible pointers rather than rewriting source records.

This architecture limits the damage caused by a bad map. It does not pretend the map is harmless; a wrong active record can still produce a wrong edge. The difference is that the lineage remains visible and the derived structure is not load-bearing for canonical history.

Important boundary. A rebuild works only from retained canonical evidence. It cannot restore source content that was deliberately deleted or lost without a backup.

10 Multi-Hop Recall That Can Explain Itself

Article 10 of 20

Some memory questions require joining facts, not merely finding a similar sentence.

Consider two accepted memories: “my boss is Sarah” and “Sarah’s preferred airport is SEA.” A question about the user’s boss and flights may share words with the first statement but not with the answer statement.

The derived graph connects these records through entities and typed relations. Retrieval starts with a small number of lexical seeds, then spreads through a bounded frontier for a limited depth. The default design uses no learned graph weights; trust, relation priors, recency, and hop decay are versioned deterministic inputs.

The useful output is not only “SEA.” It is the path that explains why the record was nominated: the query reached the boss relation, that relation identified Sarah, and Sarah’s accepted airport relation led to SEA. Every edge cites its source record.

Graph candidates and direct lexical candidates are combined with a rank-fusion method rather than pretending their scores have the same meaning. The graph remains optional, and direct record recall remains available when an item cannot be represented as an edge.

In the paper’s specific 10,002-record probe, direct top-10 recall missed the older airport record, while graph recall returned it at rank 2 after visiting two edges. This validates the constructed mechanism, not general question answering quality.

Important boundary. Path explanation proves how a result was reached, not that every relation on the path is factually correct.

11 Recall Forensics: Explaining a Past Answer

Article 11 of 20

A memory system should be able to reconstruct why a specific result appeared at a specific time.

Traditional retrieval evaluation asks whether the system found something relevant. Recall forensics asks a deployment question: if a surprising memory appeared last Tuesday, what durable evidence explains its ranking, source, lifecycle state, and integrity?

The AETNAMEM paper defines five checks. Attribution recomputes ranking inputs and ordering. Provenance follows the result back to admission, episode, and origin. Lifecycle replay checks whether the item was active at that turn. Decision replay reconstructs the logged selection. Integrity testing mutates selected evidence fields to see whether verification notices.

In the content-addressed evaluation snapshot, an independent standard-library auditor recomputed 14 of 14 ranking decisions and reconstructed all 68 returned result pairs. It detected all 75 selected single-row mutations. Eleven stable turns were re-executed exactly.

Three earlier turns were deliberately excluded from exact engine re-execution because a later correction or deletion changed their canonical inputs. That exclusion is part of the result: preserving deleted plaintext only to make an audit demo look complete would violate the deletion contract.

Recall forensics is not the same as proving an answer true. It proves what the measured evidence can reconstruct about why the system returned it.

Important boundary. The evaluated workload is synthetic and bounded. Its mutation coverage does not prove detection of every possible coherent database rewrite.

12 A Hash Chain Is Useful, but It Needs an Anchor

Article 12 of 20

Tamper evidence stored beside the data cannot, by itself, prove that the entire database was not replaced.

AETNAMEM appends important transitions to a per-subject SHA-256 hash chain. Each event commits to the previous event hash. An edit, insertion, or interior deletion breaks verification from the chain start.

This is valuable, but it has a limit. An attacker with complete database control could truncate the tail or rewrite the entire chain and calculate new hashes. If the verifier trusts only the modified file, there is no independent reference for the old head.

The answer is an external checkpoint. A checkpoint records the event count, sequence, and current hash, then moves that commitment to an independent location such as append-only object storage or a time-stamping service.

The paper's mutation experiment demonstrates the distinction directly. Tail truncation was detected in zero of five trials using the chain alone and in five of five when checked against an anchored head.

An incremental verification cache can make routine checks faster by verifying only the new suffix. That cache is an optimization, not a trust anchor, because an attacker who replaces the database may also replace a local cache.

Important boundary. A hash chain provides tamper evidence relative to a trusted point. It does not provide replication, availability after disk loss, trusted wall clock time, or automatic recovery.

13 Confidence Is Not Trust

Article 13 of 20

One number should not decide whether information is relevant, trusted, true, or authorized.

AI systems often use the word “confidence” as if it described one property. In practice, several different questions are being mixed together.

AETNAMEM separates extraction confidence, source trust, record retrieval score, graph activation, entity-merge confidence, and external-action outcome. Each value has a limited job.

Extraction confidence records how strongly a deterministic parser recognized a supported sentence form. Trust records where the statement came from and whether it was reviewed. Retrieval scores order already eligible candidates. Graph scores measure the strength of a typed path. Merge confidence describes an identity proposal. Action outcomes report what was observed after an external operation.

A high retrieval score cannot activate quarantined content. A high merge score cannot merge two entities automatically. A confident memory cannot authorize a file change. An external timeout is not converted into “60 percent successful”; it becomes uncertain because the outside state is unknown.

This separation makes the system less convenient to summarize with one dashboard number, but much easier to reason about. Every value answers a specific question and has a defined authority boundary.

Important boundary. The current ranking and merge values are deterministic heuristics, not calibrated probabilities of truth or correctness.

14 Ambiguity Should Stay Visible

Article 14 of 20

When the system cannot justify a conclusion, abstention or review is better than a hidden guess.

Language is ambiguous. Identities are ambiguous. External systems can fail in ambiguous ways. A memory architecture becomes fragile when it compresses all of these situations into a confident-looking answer.

The current AETNAMEM extractor is intentionally narrow. Unsupported phrasing creates a source episode but may create no semantic record. Questions are not treated as assertions. A caller can request a forced note, but that choice is explicit and recorded.

Unstructured conflicts may remain as multiple records rather than being silently merged. Entity-resolution candidates wait for a reviewer. Exact name or alias overlap can support a proposal, but it does not prove two entities are the same.

External ambiguity receives its own states. If an operation fails after dispatch, generic code may not know whether the provider completed it. The system records uncertainty or recovery required instead of retrying blindly and risking a duplicate effect.

This approach gives users more visible states to understand. That is a design cost, but it reflects reality. A single green check mark is not useful when the system cannot prove what happened.

Important boundary. Abstention reduces silent corruption but can reduce convenience and recall coverage. The product must make review workflows understandable rather than leaving uncertainty as an unexplained error.

15 Information Is Not Authority

Article 15 of 20

Remembering a fact should not grant permission to act on it.

An assistant may remember that a report belongs in a certain folder. That memory can help prepare a proposal. It should not, by itself, authorize the assistant to overwrite a file.

AETNAMEM separates memory from external authority. Recalled records can inform an action plan, but enforced write operations require a trusted, host-attested authority reference. Untrusted records may contribute context without becoming permission.

This distinction matters for prompt injection. A web page might tell an agent to edit a configuration file. Even if the content is stored for review, it cannot manufacture the user authority required by the guarded action engine.

The agent-facing interface can stage tools and inspect state, but it cannot label its own proposal as a user instruction or create approval material. The reviewer-side interface creates authority from the user's actual task.

This is one of the main reasons AETNAMEM is more than a retrieval component. It connects memory evidence to an effect-control plane while keeping the two roles distinct.

Important boundary. The guarantee applies only when model-requested effects are routed through the registered broker. Direct shell, filesystem, database, or provider access can bypass the gate and must be restricted by deployment policy.

16 Approve the Exact Plan, Not the General Intention

Article 16 of 20

Approval should bind to the precise operation that was reviewed.

“Update my report” is an intention, not a complete execution plan. The file may change after review. An adapter may be upgraded. A path may resolve differently. The proposed content may no longer match what the user saw.

AETNAMEM prepares a canonical plan containing the operation, arguments, preview, preconditions, adapter manifest, dependencies, evidence references, and policy version. It calculates a digest over that complete plan.

Reviewer approval uses HMAC-SHA256 to bind the transaction identifier, plan digest, approver label, issue time, expiry time, and a random nonce. The approval applies to one exact plan for a limited period, not to future variations of the same idea.

Before execution, the engine recomputes the plan and approval commitments, checks the audit chain, confirms the adapter did not change, and revalidates preconditions. Each operation is checked again immediately before its effect.

For the reference filesystem adapter, a changed file digest causes the plan to stop instead of stretching the old approval to cover new state. After a write, the adapter independently reads the result and verifies its expected digest.

Important boundary. An HMAC proves possession of a shared secret, not a person’s legal identity. Multi-user or regulated deployment would require stronger identity, revocation, key custody, and policy administration.

17 “Uncertain” Is a Responsible Result

Article 17 of 20

When an external call may have succeeded, pretending failure and retrying can be more dangerous than admitting uncertainty.

Database operations can often commit or roll back atomically. External calls cannot. A provider may receive a request, perform the effect, and disconnect before returning a response. The caller sees an error but does not know the outside state.

AETNAMEM persists execution state before dispatch and does not hold a SQLite transaction open across the external call. If the call raises after dispatch, the action becomes uncertain. If the process restarts during a committing or compensating operation, the transaction becomes recovery required.

Provider-specific idempotency lookup or authoritative state inspection may later resolve the outcome. Generic code does not retry blindly, because a second request could duplicate the first effect.

For multi-step plans, previously verified effects may be compensated in reverse order. Compensation runs only when current state still matches the engine’s verified after-state. If another actor changed the resource, the engine refuses to overwrite that later work.

The final result may be committed, compensated, partial, uncertain, or recovery required. These states are less tidy than success and failure, but they preserve what the system actually knows.

Important boundary. A generic control plane cannot resolve ambiguity when a provider offers neither idempotency records nor an authoritative way to inspect the resulting state.

18 What Aetnamem Encryption Protects Today

Article 18 of 20

Encryption claims are meaningful only when they name the platform, files, process state, and remaining plaintext.

The current AETNAMEM release does not use SQLCipher and does not provide one uniform live-database encryption design across macOS, Windows, and Linux.

On macOS, the desktop launcher can seal an idle database using OpenSSL AES-256-CBC with PBKDF2 and an HMAC-SHA256 sidecar. The secret is held in the login Keychain. Before opening the database, the HMAC is checked; during use, SQLite operates on a plaintext database in a temporary directory.

The launcher checkpoints and reseals after mutations, subject to a short rate limit, and force-seals during a clean shutdown. This provides useful confidentiality when an attacker obtains only the idle sealed files without the Keychain secret.

It is not transparent database encryption. The live database and WAL are plaintext while the service runs. The ciphertext and HMAC sidecar are not published atomically. The same secret currently serves two cryptographic roles, and archive or workspace files need separate protection.

Windows and Linux currently store ordinary SQLite files. Their at-rest confidentiality depends on operating-system controls such as BitLocker or LUKS when those controls are enabled and managed correctly.

Important boundary. Encryption protects data against specific storage threats. It does not protect a database from every process running as the unlocked user, prove memory correctness, or replace tested backup and key-recovery procedures.

19 Scale Claims Should Follow Evidence

Article 19 of 20

An architecture can be designed for growth without pretending that unmeasured capacity has already been proven.

AETNAMEM uses three mechanisms to avoid obvious unbounded work in hot recall: candidate-capped direct ranking, bounded graph traversal, and optional cold partitions for inactive derived edges. Incremental audit verification avoids rehashing an already validated prefix on every routine check.

These choices make larger memory stores plausible. They do not establish a supported maximum. SQLite file size, full-text selectivity, cache pressure, audit growth, backup duration, archive churn, concurrent writes, and hardware all affect real deployments.

The paper therefore reports what it measured. In one Apple M2 probe with 10,002 records, direct recall had a median of 17.139 milliseconds and graph recall 24.799 milliseconds. Graph recall found the constructed two-hop target at rank 2; direct top-10 recall did not. The candidate cap remained 200.

That experiment supports the intended bounded-recall and multi-hop mechanism for one synthetic scenario. It is not a million-record benchmark, a multilingual relevance evaluation, a sustained concurrency test, or evidence of state-of-the-art answer quality.

Engineering credibility improves when design projections and validated results remain separate. The next useful evidence is not a larger adjective; it is a broader workload, reproducible measurements, and published failure conditions.

Important boundary. Performance numbers vary by host and run. The more reproducible claims are the configured bounds, returned rank, graph path, and visited-edge count.

20 The End Goal: Governed Memory Infrastructure

Article 20 of 20

The long-term value is a memory layer that remains useful as models, interfaces, and applications change.

AETNAMEM is designed so that memory does not belong to one language model. The model can be local or remote and can change over time. Canonical records, trust transitions, graph lineage, audit evidence, and approvals remain outside the model.

This makes the system potentially useful in several roles: personal assistant memory, memory for agents using MCP or an SDK, governed intake from external channels, and decision memory around engineering workflows. The application may use recalled records to augment a prompt, but retrieval augmentation is only one integration point.

The more precise description is a **governed memory and effect-control plane**. It keeps accepted information durable, separates suspicious material, shows why a result surfaced, distinguishes information from authority, and records uncertainty when an external outcome cannot be proven.

The architecture does not depend on claims that every component is new. Full text search, graphs, provenance, hash chains, approvals, and compensating actions are established techniques. The systems contribution is their composition around inspectable memory transitions and approval-bound effects.

The remaining work is concrete: stronger cross-platform encryption, broader retrieval-quality evaluation, larger scale tests, operating-system sandboxing, better adapters, and independent security review. Those improvements matter more than presenting the system as finished.

The intended user experience is simple: remember what was accepted, show where it came from, keep unreviewed content separate, ask before changing the world, and admit when the evidence is incomplete.

Important boundary. AETNAMEM is an evolving open-source system. Its potential should be judged against implemented behavior, reproducible evidence, and explicit limitations rather than unsupported claims of universal novelty.